

Representing Non-numeric Data

CSCI 1101 Introduction to Computer Science

Representing Text

What must be provided to represent text?

There are finite number of characters to represent, so list them all and assign each a binary string

Character set

A list of characters and the codes used to represent each one

Computer manufacturers agreed to standardize

The ASCII Character Set

ASCII stands for American Standard Code for Information Interchange

ASCII originally used seven bits to represent each character, allowing for 128 unique characters

Later **extended ASCII** evolved so that all eight bits were used

How many characters could be represented?

ASCII Character Set Mapping

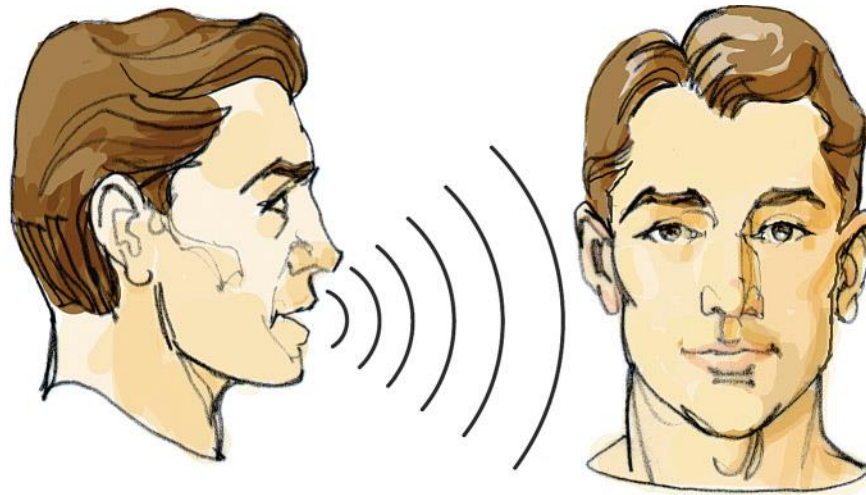
Left Digit(s)	Right Digit	ASCII									
		0	1	2	3	4	5	6	7	8	9
0		NUL	SOH	STX	ETX	EOT	ENQ	ACK	BEL	BS	HT
1		LF	VT	FF	CR	SO	SI	DLE	DC1	DC2	DC3
2		DC4	NAK	SYN	ETB	CAN	EM	SUB	ESC	FS	GS
3		RS	US	□	!	“	#	\$	%	&	'
4		(	)	*	+	,	-	.	/	0	1
5		2	3	4	5	6	7	8	9	:	;
6		<	=	>	?	@	A	B	C	D	E
7		F	G	H	I	J	K	L	M	N	O
8		P	Q	R	S	T	U	V	W	X	Y
9		Z	[	\	]	^	_	`	a	b	c
10		d	e	f	g	h	i	j	k	l	m
11		n	o	p	q	r	s	t	u	v	w
12		x	y	z	{		}	~	DEL		

The Unicode Character Set

Code (Hex)	Character	Source
0041	A	English (Latin)
042F	Я	Russian (Cyrillic)
0E09	๒	Thai
13EA	Ꮝ	Cherokee
211E	℞	Letterlike Symbols
21CC	⇒	Arrows
282F	⠋	Braille
345F	佤	Chinese/Japanese/ Korean (Common)

Figure 3.6 A few characters in the Unicode character set

Representing Audio Information



We perceive sound when a series of air compressions vibrate a membrane in our ear, which sends signals to our brain

Representing Audio Information

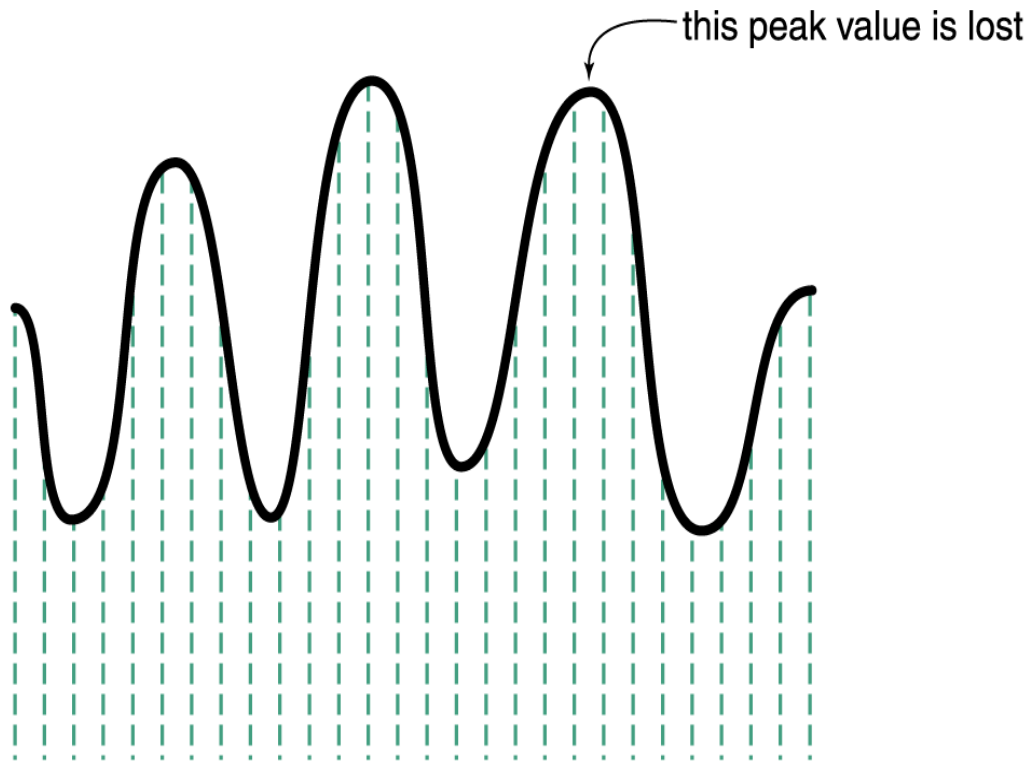
Digitize the signal by sampling

- periodically measure the voltage
- record the numeric value

How often should we sample?

A sampling rate of about 40,000 times per second is enough to create a reasonable sound reproduction

Representing Audio Information



Some data
are lost, but a
reasonable
sound is
reproduced

Figure 3.8 Sampling an audio signal

Representing Images and Graphics

- Color is expressed as an RGB (**red-green-blue**) value
 - These three numbers indicate the relative contribution of each of these three primary colors **of light**
 - Each color value is represented as one byte, so each color value ranges from 0 to 255
 - This 3-byte color is called **true color**.

Representing Images and Graphics

RGB example:

An RGB value of (255, 255, 0) maximizes the contribution of red and green, and minimizes the contribution of blue, which results in a bright yellow

Representing Images and Graphics

Remember that we are talking about **additive color mixing (mixing light)**... (used in displays)

... not **subtractive color mixing (mixing pigments)** (used in printers). Example, **CMYK**

Representing Images and Graphics

Do we really “need” all those bits?

- Number of colors we can represent with 24- bit true color = $2^{24} = 16.7+$ million
- Number of colors humans can perceive = ?
 - Low end: 100,000
 - High end: 10 million

Digitized Images and Graphics

Digitizing a picture

Representing it as a collection of individual dots called pixels

Resolution

The number of pixels used to represent a picture

Raster Graphics

Storage of data on a pixel-by-pixel basis

Bitmap (BMP), GIF, JPEG, and PNG are raster-graphics formats

Digitized Images and Graphics

Bitmap format

Contains the pixel color values of the image from left to right and from top to bottom

GIF format (indexed color)

Each image is made up of only 256 colors

JPEG format

Averages color hues over short distances

PNG format

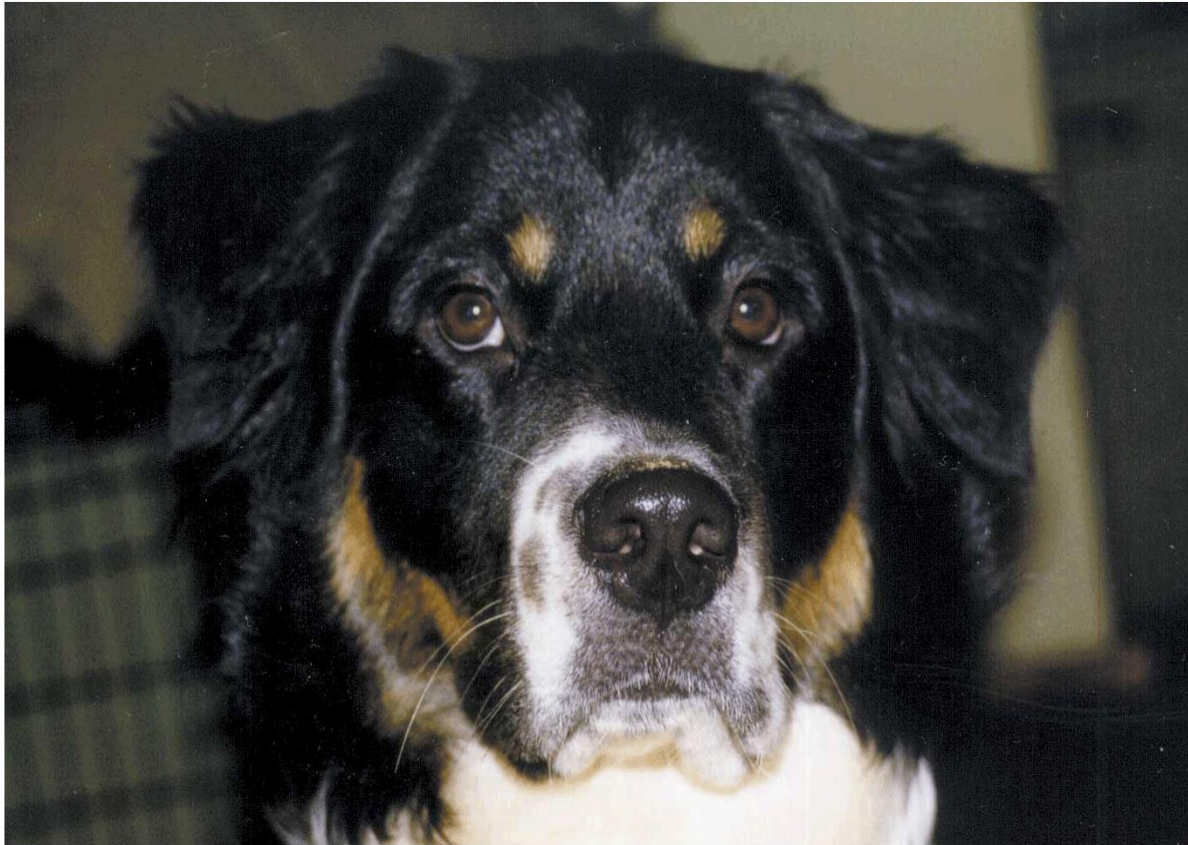
Like GIF but achieves greater compression with wider range of color depths

Which is better for line drawings? Pictures?

For more information about formats check:

<http://www.scantips.com/basics09.html>

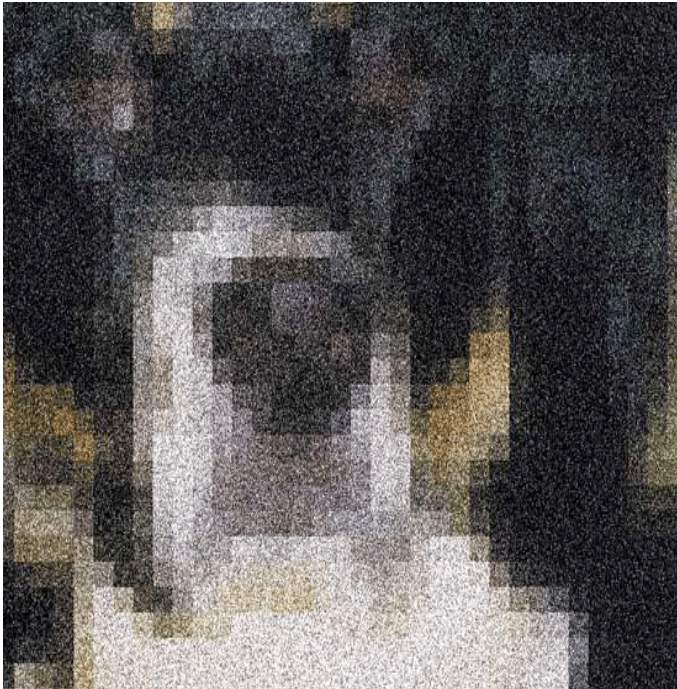
Digitized Images and Graphics



Whole
picture

Figure 3.12 A digitized picture composed of many individual pixels

Digitized Images and Graphics



Magnified portion
of the picture

See the pixels?

Figure 3.12 A digitized picture composed of many individual pixels

Vector Graphics

Vector graphics

A format that describes an image in terms of lines and geometric shapes

A vector graphic is a series of commands that describe a line's direction, thickness, and color

The file sizes tend to be smaller because not every pixel is described

Vector Graphics

The good side and the bad side...

Vector graphics can be resized mathematically and changes can be calculated dynamically as needed

Vector graphics are *not* good for representing real-world images

Representing Video

Video codec COnpressor/DECompressor

Methods used to shrink the size of a movie to allow it to be played on a computer or over a network

Almost all video codecs use lossy compressions to minimize the huge amounts of data associated with video